

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

- Remove the front cover. Refer to Procedure 001-031 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-031-tr.html).
- Rotate the engine by turning the crankshaft until the index marks on the crankshaft and camshaft gears are in alignment.

Note : The marks on the crankshaft and camshaft gears **must** point straight at one another. The marks on the idler gear do not have to be aligned as long as the crankshaft and camshaft gears are in alignment.



Remove

Bolt-On Type

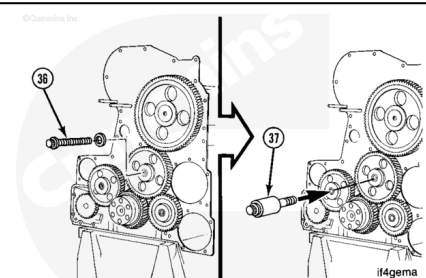
The bolt-on idler shaft has a flange that requires the shaft, gear, and thrust washers to be removed as an assembly.

Note : The shafts used in newer engines do not have threads to be used with a puller. Loosen the capscrew (36), but do not completely remove it.

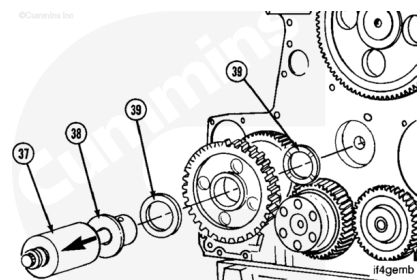
Remove the capscrew (36).

Use an old piston pin and a K19 cylinder head capscrew as a slide hammer (37).

Thread the capscrew into the idler shaft.

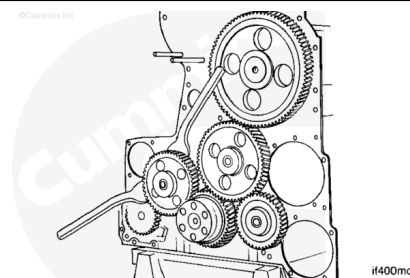


Use the slide hammer (37) to remove the shaft (38), two thrust washers (39), and the gear as an assembly.



For shafts used in newer engines, make sure the capscrew is threaded into the cylinder block.

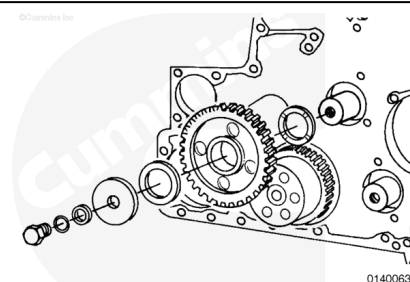
Pry the gear and shaft assembly out of the cylinder block with two pry bars.



Press-Fit Type

Some engines have a press-fit design idler shaft that does **not** have to be removed from the block to remove the idler gear.

Remove the capscrew, washers, gear, and thrust washer.

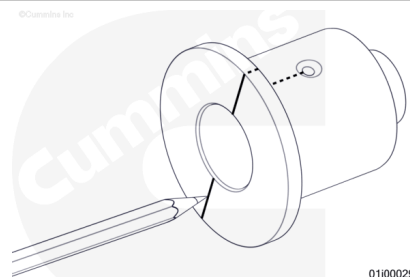


Install

Bolt-On Type

⚠ CAUTION ⚠

The oil holes in the camshaft idler shaft and hydraulic pump idler shaft must be installed at a specific orientation. If the shaft is not oriented correctly, a malfunction of the gear bushing can result.

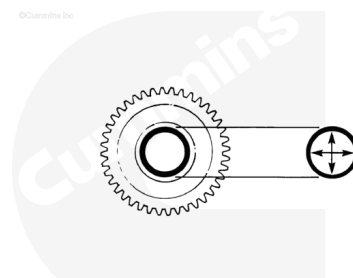


Mark the flange of the shaft to show the oil hole orientation.

Measure the inside diameter.

Idler Gear Bushing Inside Diameter

mm		in
47.638	MIN	1.8755
47.714	MAX	1.8785

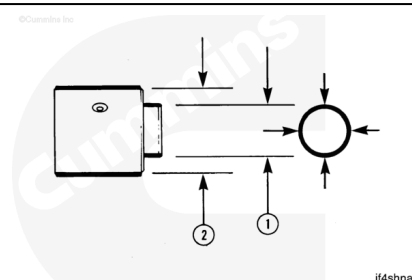


Note : The bushing in the gear is precision-bored after installation. If machining capability is not available, replace the bushing and the gear as an assembly.

Measure the outside diameter.

Camshaft Idler Shaft Outside Diameter (1)

mm		in
25.387	MIN	0.9995
25.400	MAX	1.0000



Camshaft Idler Shaft Outside Diameter (2)

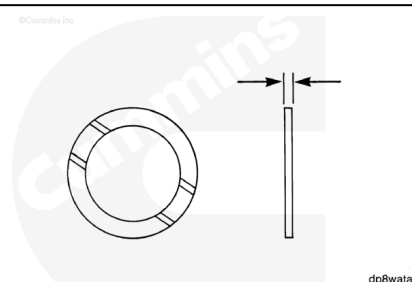
mm		in
47.549	MIN	1.8720
47.600	MAX	1.8740

Check the grooved side for damage.

Measure the thrust bearing thickness.

Thrust Bearing Thickness

mm		in
0.235	MIN	0.092
0.245	MAX	0.096



Oversize thrust bearings are available to adjust the end clearance.

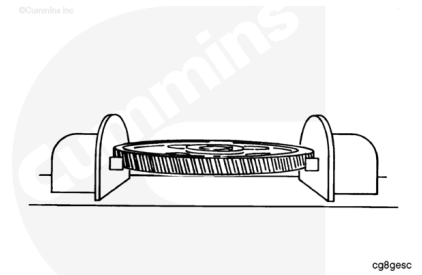
Magnetic Crack Inspect

⚠ CAUTION ⚠

Use copper braid contact that has neoprene covers to avoid burning the teeth of the gear.

⚠ CAUTION ⚠

If the gear contains a keyway, position the gear so that the keyway points toward one of the contacts when checking.



Use a magnetic particle testing machine.

Use the residual method. Apply head shot amperage.

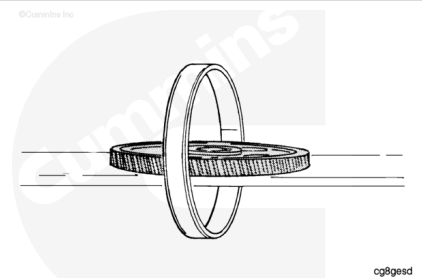
Adjust the amperage to the specified value.

Gear Outside Diameter	Amperage DC
Less than 101 mm [4.0 in]	1000
101 to 203 mm [4.0 to 8.0 in]	1500
Greater than 203 mm [8.0 in]	2000

Check the gear for cracks.

⚠ CAUTION ⚠

If the gear contains a keyway, position the gear so that the keyway points toward one of the contacts when checking.



Use a magnetic particle testing machine.

Use the residual method. Apply head shot amperage.

Adjust the amperage to the specified value.

Gear Outside Diameter	Amperage DC
Less than 101 mm [4.0 in]	4000
101 to 203 mm [4.0 to 8.0 in]	6000
Greater than 203 mm [8.0 in]	8000

Note : Ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

Check the gear for cracks.

Limits of Acceptance - Machined Surfaces

An open indication is visible to the eye after the wetting operation has been completed. An indication below the surface is **not** visible to the eye after the wetting operation has been completed. An indication below the surface can be seen with the use of ultraviolet light that is

part of the machine.

Do **not** use the gear if:

- There is an open indication.
- There is an indication below the surface that is in the shaded area as shown.
- There is an indication below the surface that is longer than 6.0 mm [0.24 in].

Limits of Acceptance - Forged Surfaces

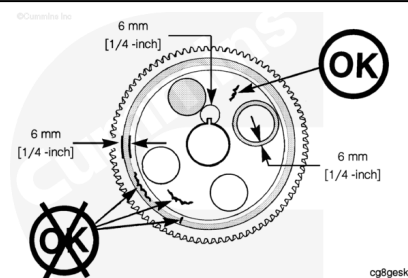
Do **not** use the gear if:

- There is an open indication that is in a circumferential direction.
- There is an open indication that is longer than 9.5 mm [0.374 in].
- There is an indication below the surface that is in the shaded area.

An indication below the surface in a radial direction is acceptable (OK) in the area that is **not** shaded.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ CAUTION ⚠

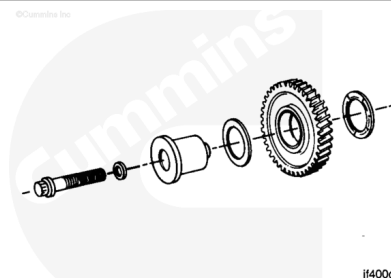
The gear must be demagnetized completely and cleaned thoroughly. Any small metal particles will cause engine damage.

Demagnetize the gear.

Use solvent or steam to clean the part.

⚠ CAUTION ⚠

To reduce the possibility of engine damage, the grooves in the thrust washers must be positioned toward the gear.



The timing marks on the camshaft idler gear must be visible when the gear is installed.

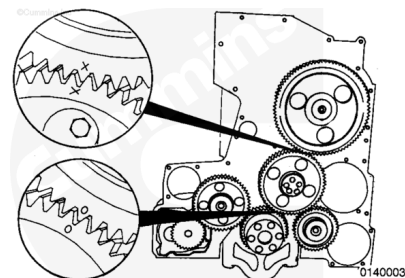
Lubriplate® Number 105 multipurpose lubricant, Part Number 3163086, or equivalent.

Lubricate the capscrew with clean engine oil.

Assemble the parts as illustrated in the graphic.

⚠ CAUTION ⚠

Do not allow the inner thrust washer to fall between the shaft and block or the washer will be damaged and the gear end clearance will be too large.

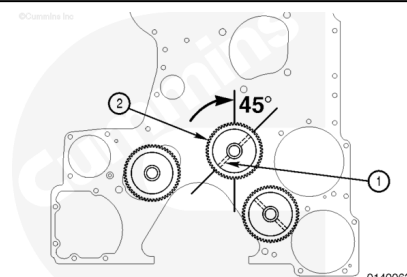


The camshaft idler gear is the only gear that has timing marks.

Install the gear and shaft assembly aligning the following:

- The "O" on the idler gear with the "O" on the crankshaft gear.
- The "X" on the idler gear with the "X" on the camshaft gear.

Align the camshaft idler shaft (2) so the oil drilling shaft (1) is 45 degrees **clockwise** from vertical as shown.

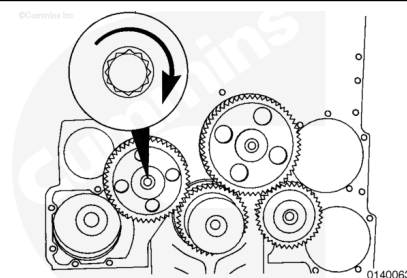


Install the capscrew and use it to pull the shaft into the bore.

Tighten the capscrew.

Torque Value:

1. 185 n•m [135 ft-lb]
2. Loosen completely.
3. 60 n•m [45 ft-lb]
4. Tighten 90 degrees.

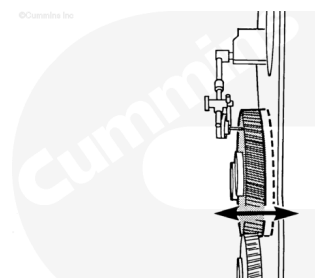


Measure the camshaft idler gear end clearance.

Camshaft Idler Gear End Clearance		
mm		in
0.10	MIN	0.004

Camshaft Idler Gear End Clearance

mm		in
0.36	MAX	0.014



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If the clearance is **not** within specifications, check for foreign material between the parts, or check for proper location of the thrust washers.

Oversize thrust washers are available.

Press-Fit Type

⚠ CAUTION ⚠

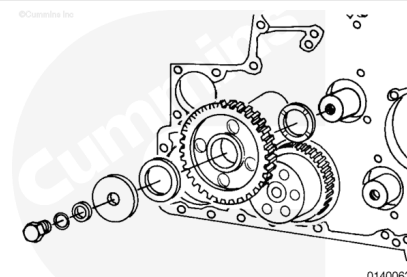
To reduce the possibility of engine damage, the grooves in the thrust washer must be positioned toward the gear.

Lubricate the gear bushing, shaft, and thrust washers with Lubriplate® Number 105 multipurpose lubricant, Part Number 3163086, or equivalent.

Lubricate the capscrews with clean engine oil.

Assemble the parts as illustrated in the graphic.

Torque Value: 45 N·m [33 ft-lb]



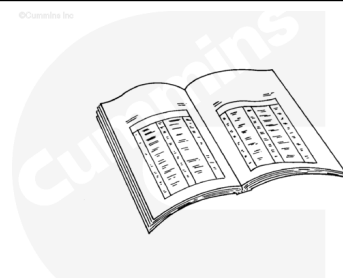
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Finishing Steps

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

- Install the front gear cover. Refer to Procedure 001-031 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-031-tr.html).



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